# Component 3 Data Strategy (FishGuard AI)

**Objective:** Develop a machine learning model to forecast tuna availability/catch using fisheries records and environmental indicators.

## STEP 1 - DATA SOURCE CLASSIFICATION

During the research, multiple fisheries-related data sources were identified. They are classified into two categories:

### Category A - Core Modeling Data

*These datasets directly contribute to model training.*

**Fisheries Logbook Data (Primary Fisheries Dataset)**

**Contains information such as:**

- Species Name
- Weight
- Quantity
- Date
- Vessel ID
- Gear Type
- Fishing Location
- Ports
- Coordinates

**Reason for selection:**

- Provides detailed fishing activity records.
- Contains spatial and temporal information.
- Allows integration with environmental datasets.
- Supports tuna-specific filtering.
- Suitable for machine learning.

**Status:** Main fisheries dataset.

### Category B - Reference & Supporting Data

**Ministry of Fisheries Statistics**

**Sources:** Monthly Fish Production, Annual Fish Production, Fisheries Statistical Bulletins, Fleet Statistics, Infrastructure Statistics

**Purpose:** Background information, Trend analysis, Research discussion, Result validation, Viva evidence

*Not intended as a primary machine learning dataset.*

**Reason:** Data is aggregated. Limited species-level information. Limited spatial information. Cannot directly link catches to environmental conditions.

**Status:** Supporting dataset.

**Export / Import Statistics**

**Contains:** Tuna exports, Export quantities, Export values, Destination countries

**Purpose:** Industry context, Market discussion, Future work, Economic interpretation

*Not directly used for forecasting tuna availability.*

**Status:** Reference only.

## STEP 2 - DATA STORAGE STRATEGY

**GitHub Repository**

Store only: Code, Notebooks, Processed datasets, Documentation, README

*Repository should remain clean.*

**OneDrive**

Store: Ministry PDFs, Official Reports, Export/Import Files, Annual Statistics, Raw Environmental Data, Large Original Files

**Suggested structure:**

## STEP 3 - ENVIRONMENTAL DATA DECISION

**Selected:** Chlorophyll-a (CHL)

**Reason:**

- CHL is a standard indicator of ocean productivity.
- Frequently used in fisheries forecasting studies.
- Provides a single representative measure.
- Reduces feature complexity.

**Not selected initially:** DIATO, DINO, GREEN, HAPTO, MICRO, NANO, PICO, PROCHLO

**Reason:**

- These represent individual phytoplankton groups.
- More specialized variables.
- May introduce redundancy.
- Can be explored in future feature engineering.

## STEP 4 - FEATURE SELECTION STRATEGY

**Important principle:** *Data Selection ≠ Feature Selection*

### Data Selection

Choosing which datasets to collect.

*Example:* Logbooks, Chlorophyll, SST, Rainfall

### Feature Selection

Choosing which fields enter the model.

*Example:* CHL, SST, Month, Catch, Effort, CPUE

## STEP 5 - ANSWER FOR VIVA

### How were variables selected?

Variables were selected based on fisheries research literature, domain knowledge, and relevance to tuna availability. After model development, feature importance and Explainable AI methods such as SHAP were used to evaluate the contribution of each feature.

### Why not use all available variables?

Not every available variable contributes meaningful predictive information. Feature selection was performed to reduce redundancy, improve interpretability, and focus on variables with a plausible relationship to tuna availability.

### Where did the fisheries data come from?

*Honest answer:* Fisheries logbook records were obtained through academic collaboration and subsequently validated against official fisheries statistics, species information, and ministry publications before being considered for the research.

## STEP 6 - CURRENT PROJECT DECISION

**At this moment:**

✅ Chlorophyll-a source identified
✅ Ministry statistics organized
✅ Export/Import statistics archived
✅ Data storage strategy decided
✅ Fisheries logbook dataset identified as primary candidate
✅ Ready for fisheries column analysis

## NEXT STAGE